Quiz 4: TCP Congestion Control (graded)

In class, Wednesday November 18, 2026. 20 minutes. Closed book.

- Q1. A TCP Reno sender has cwnd = 16 MSS and ssthresh = 32 MSS when it receives three duplicate ACKs. What are cwnd and ssthresh immediately afterwards?
- Q2. A TCP sender in slow start begins with cwnd = 1 MSS. How many round-trip times does it take to reach a cwnd of 64 MSS, assuming no loss?
- Q3. With MSS = 1460 bytes, RTT = 100 ms and a loss rate of 1%, estimate the average TCP throughput in Mbit/s.
- Q4. Explain in two sentences why AIMD converges to fairness between two competing flows.